

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	30 June 2026
Team ID	
Project Name	Cosmetic Insights: A Tableau-Based Analysis of Skincare Trends and Consumer Preferences
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the architectural flow (see Solution Architecture document) and the information per Table-1 and Table-2 below.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	Data Source	Raw cosmetics product data file	CSV (cosmetics.csv)
2	Data Cleaning	Standardizing category labels, verifying skin-type flags, removing inconsistent rows	Microsoft Excel
3	Data Source Connector	Connects cleaned data to the visualization layer	Tableau Data Source ('cosmetics')
4	Visualization / Application Logic	Worksheet creation, aggregation, filtering logic for all 18 worksheets	Tableau Desktop / Tableau Public
5	Presentation Layer	Guided narrative combining key worksheets	Tableau Story ('Story 1')
6	Workbook Packaging	Bundles data extract with workbook for sharing	Tableau Packaged Workbook (.twbx)
7	File Storage	Storage of dataset, workbook, and documentation files	Local filesystem / GitHub repository
8	Distribution	Publishing and sharing the final dashboard	Tableau Public
9	Documentation	Project documentation deliverables	Microsoft Word / PDF
10	Version Control & Submission	Hosting the final project folder structure for internship submission	GitHub

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No paid/licensed software required beyond standard institutional access	Tableau Public (free), Microsoft Excel
2	Security Implementations	No personally identifiable information in the dataset; access limited to team and published link	N/A — public product dataset
3	Scalable Architecture	Single federated data source feeding all worksheets; new rows can be added without redesigning charts	Tableau extract-based data source
4	Availability	Packaged .twbx workbook and Tableau Public link remain accessible independent of original raw files	Tableau Public hosting
5	Performance	Dataset size (~1,500 rows) is small enough for all worksheets to render in a few seconds without caching/CDN	Tableau in-memory extract

References:

Reference
https://www.tableau.com/products/public
https://www.tableau.com/learn/articles/data-visualization